

Cancer Rates among American Indians and Alaska Natives

Is there a National Perspective?

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BACKGROUND. Two important goals of cancer surveillance are to provide milestones in the effort to reduce the cancer burden and to generate observations that form the basis for cancer research and intervention for cancer prevention and control. Determination of the cancer burden among American Indians and Alaska Natives (AIAN) has been difficult largely due to lack of data collection efforts in many areas of the country and misclassification of racial data that results in undercounting of Native Americans. There is a revitalized commitment to improve data collection among the national agencies and organizations.

METHODS. Data on cancer trends from 12 areas covered by the Surveillance, Epidemiology and End Results (SEER) Program were reviewed for incidence and death rates for 1992–2000. AIAN trends were examined and compared with trends among other racial/ethnic population groups. Reference was made to studies of disease-specific survival for nine of the SEER areas for 1988–1997.

RESULTS. In SEER areas, cancer incidence rates for AIAN populations appeared to be decreasing significantly for lung and breast cancers among women and for prostate cancer among men. However, death rates rose, although not significantly, over the same period, except for a significant decrease in prostate cancer. Among the cancers with rising death rates were lung cancer (AIAN women) and colorectal cancer (AIAN men). In addition, survival often was lower for AIAN populations.

CONCLUSIONS. Although the incidence was stable or decreased among AIAN populations, increased death rates and lower survival rates indicate the need for intensified application of cancer prevention and control measures, including screening and treatment. Difficulties in interpretation of data include small population size and substantial interregional differences in rates. *Cancer* 2003;98:1262–72. Published 2003 by the American Cancer Society.*

KEYWORDS: cancer surveillance, health disparities, cancer rates, American Indian, Alaska Native, cancer trends, cancer incidence, cancer mortality, cancer death rate, data collection.

Quantitative by definition, cancer surveillance provides a portrait of cancer and its determinants in specified populations. Its core functions are not only measurement of incidence, death, and survival, but also assessment of genetic disposition, environmental and behavioral risk factors, screening practices, and quality of care from prevention through diagnosis, treatment, and palliation. In addition, it can incorporate the social and economic context of cancer as a basis for policy decision-making. Two important uses of cancer surveillance are 1) to provide milestones in the effort to reduce the cancer burden and 2) to generate observations that form the basis for cancer research and intervention for cancer

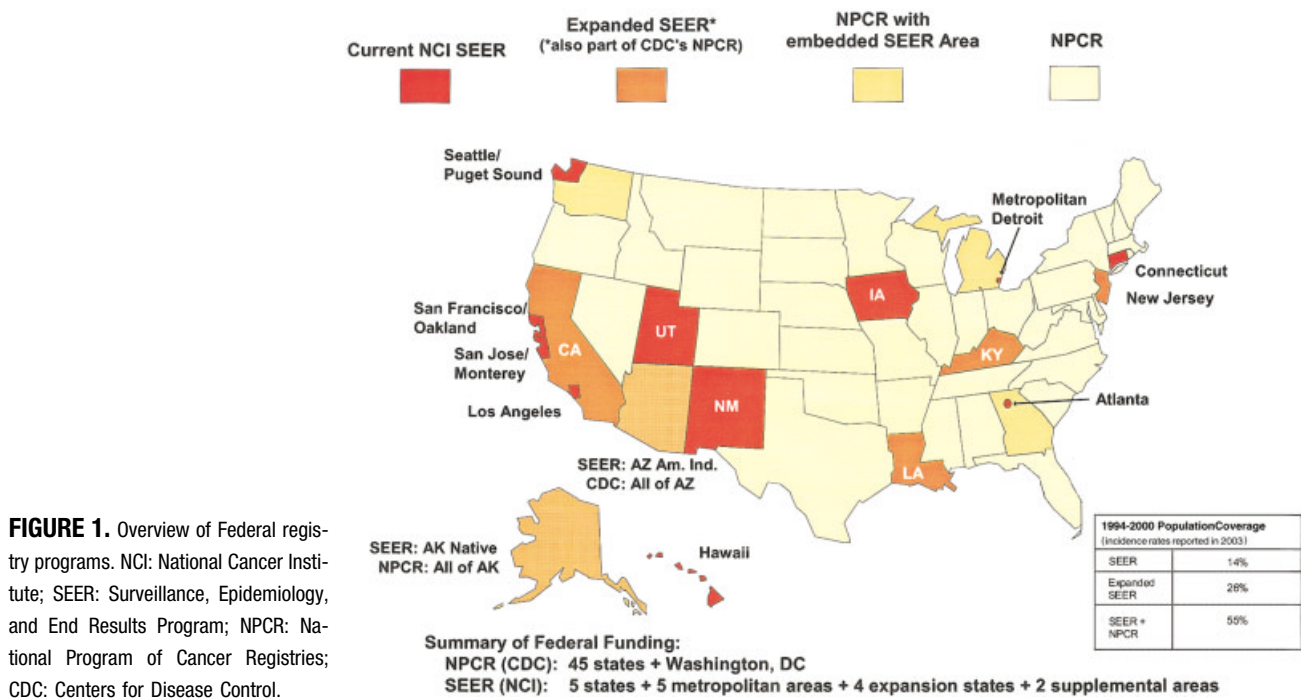


FIGURE 1. Overview of Federal registry programs. NCI: National Cancer Institute; SEER: Surveillance, Epidemiology, and End Results Program; NPCR: National Program of Cancer Registries; CDC: Centers for Disease Control.

prevention and control.¹ For the purposes of this article, we asked a general, two-part question: What is the level of cancer in the population, and where are the trends leading? The specific objective was to determine, as well as we could from available data, how American Indians and Alaska Natives (AIAN) are faring compared with others.

The American Cancer Society estimates that in 2003, there will be 1.33 million new cases and more than 550,000 deaths from cancer overall in the United States population.² Before describing the documented cancer rates for the AIAN population, it is important to note that complete data on AIAN do not exist. It is only in recent years that enough information has become available to create an overview. Indeed, if complete data were available from every state, it would remain difficult to describe overall AIAN cancer rates. As with most population groups, AIAN populations are not homogeneous, and the variability among tribes can be so great that an *average* would not provide a true picture. This is complicated by the small numbers available for statistical analysis, the smallest of any nationally reported group. Does the answer lie in more local data collection and reporting to provide information that is useful locally? In this article, we describe some of these efforts as well as improvements to the national systems through expansion of and collaborations among central cancer registries.

MATERIALS AND METHODS

The data presented here are from the National Cancer Institute's (NCI) Surveillance, Epidemiology, and End Results (SEER) Program, an authoritative source of information on cancer incidence and survival in the United States. Figure 1 provides an overview of national coverage by the population-based cancer registries of the SEER Program and the National Program of Cancer Registries (NPCR) of the Centers for Disease Control and Prevention (CDC). Until its expansion in 2000, SEER reported incidence and survival from 12 registries in the States of Connecticut, Iowa, New Mexico, Utah, and Hawaii and the metropolitan areas of Detroit, San Francisco—Oakland, San Jose—Monterey, Los Angeles, Atlanta (plus 10 rural Georgia counties), and the 13-county Seattle—Puget Sound area. The Alaska Native Tumor Registry, which has been collecting data since 1969, became an official member registry in 1999, with reporting beginning in 2002.

The data on incidence presented in the current article were taken from the SEER data base, which was updated most recently for the annual Cancer Statistics Review (CSR) in the spring of 2003 to include registry data through 2000, at which time SEER covered just over 30% of the AIAN population.³ The data have not been adjusted for reporting delay.⁴ For the period 1992–2000, there were 6454 AIAN cancer cases reported in the 12 SEER areas. The distribution of cases

denotes the extra efforts made to capture data on AIAN populations in 2 areas, with Alaska reporting 35% of the total and New Mexico reporting 31%. These are followed by Seattle, with almost 19%, and the combined California registries, with 6%, then Detroit and Utah, each with 2–3% of the AIAN cases reported.

Based on these numbers, it is worth offering a word of caution about interpreting rates and trends for the AIAN population. Although the cancer rates for the AIAN population generally have been reported as lower than the total population of the United States, the rates from several SEER registries are considerably higher compared with the overall SEER rates by 2-fold to 5-fold. Although not they are presented in this analysis, the rates can vary considerably between registries in different geographic areas, particularly for lung/bronchus and colorectal cancers, underlining these regional differences. This is one of the indications that differences in case ascertainment play a role in the overall rates, although other factors, such as differences in risk factors among regions of the country, also contribute to the complexity of AIAN cancer rates. At the same time, the trend data also are dominated by the registries with greatest ascertainment. It should be noted that SEER does not routinely receive information from the Indian Health Service (IHS), tribal health departments, or other AIAN health programs. The first steps being taken to remedy this are described below (see Discussion). Death rate data come from the National Vital Statistics System of the CDC National Center for Health Statistics,⁵ from which cancer deaths are coded and then reported in the CSR. The rates, as taken from the CSR, have been age-adjusted to the 2000 United States standard million population. The estimated annual percentage change in rate is calculated by first fitting a regression line to the natural logarithm of the rates using calendar year as a regressor variable. The method of weighted least squares is used to calculate the regression equation, and the estimate is tested for a significant difference from zero using the *t* distribution.⁶

Both incidence rates and death rates are calculated by dividing the number of cancer cases or deaths by a census-generated denominator; thus, rates may be influenced heavily by changes or uncertainties in census counts. The 2000 Census demonstrated that the intercensal population estimates for AIAN were underestimated throughout the 1990s. Although the counts of cases/deaths have improved and increased the numerators, the population counts in the denominators have increased also, altering some of our previous impressions of cancer rates, particularly among elderly racial/ethnic minorities.⁷ Further information on how the Census has affected cancer rates

is available at <http://www.cancer.gov/newscenter/pressreleases/Census2000>.

For trends, we focus on four major cancers for each gender for the years 1992–2000, the most recent period for which data are available on the reported racial/ethnic groups. Trends are illustrated with graphs using a statistical model that chooses line segments that change slope at *joinpoints* denoting a statistically significant change in trend.⁸ Joinpoint regression models on the natural logarithms of the rates describe the trends by a sequence of connected straight line segments. The Windows-based statistical software takes trend data and fits the simplest joinpoint model that the data allow, starting with the minimum number of joinpoints (e.g., 0 joinpoints, which is a straight line), and tests whether more joinpoints are statistically significant and must be added to the model, up to the maximum number specified by the user. This enables the user to test whether an apparent change in trend is statistically significant using a Monte Carlo permutation method. A very useful way to make sense of rates in large populations, this method has the effect of flattening trends among population groups with small numbers. With the inherent variability in annual data points for small populations, it is difficult to delineate discreet trends, and this is demonstrated with the AIAN data. In addition, joinpoints are altered by the specific years selected for analysis, because the influence of data from the year previous to the initial data point may change the slope or contribute to a change in trend direction.

SEER collects data on survival from its registries, and a recent study conducted by the NCI analyzed survival among six racial/ethnic groups for first diagnosis from 1975 to 1997 with follow-up through 1998.⁹ That analysis used the longest time frame for which data were available from the nine SEER areas that collected data over the period. Relative risks of cancer death and cancer-specific survival for more than 1.78 million cancer patients were analyzed for 6 racial/ethnic groups. The smoothed model of the survival curves are presented here for comparison of the AIAN population with other groups. The analysis of survival can be particularly complex, but it provides another context in which to compare rates.

RESULTS

For Americans as a whole, the age-adjusted cancer rates, as seen in Table 1, showed a decrease over time from 1992 to 2000 of 0.7% annually for incidence rates and 1.0% annually for death rates. Among AIAN populations, cancer incidence rates decreased from 1992 to 2000, with an estimated annual percentage change of – 3.1%. The death rate presents another story, with

TABLE 1
Cancer Incidence and Death Rates (per 100,000 population) and Estimated Annual Percent Change: Average Annual Population, Entire United States versus American Indians and Alaska Natives, 1992–2000, Age Adjusted to the 2000 Standard Population

SEER incidence ^a	Total U.S. population	Total AIAN	Total U.S. men	AIAN men	Total U.S. women	AIAN women
Incidence rate						
All sites combined	472.3	239.6	555.8	259.0	417.9	229.2
APC ^b	−0.7 ^c	−3.1 ^c	−1.7 ^c	−4.8 ^c	+0.2	−1.3
Lung and bronchus	62.6	33.1	80.8	45.6	49.6	23.4
APC	−1.2 ^c	−4.9 ^c	−2.3 ^c	−5.4	0.0	−4.1 ^c
Colon/rectum	54.2	34.7	64.2	37.5	46.7	32.6
APC	−0.6	−1.9	−0.9 ^c	−3.4	−0.3	−0.2
Prostate			170.1	53.6		
APC			−3.1 ^c	−9.8 ^c		
Breast					135.0	58.0
APC					+0.8 ^c	−3.7 ^c
Death rate^d						
All sites combined	202.3	138.0	255.5	172.3	168.3	115.8
APC	−1.0 ^c	−0.2	−1.4 ^c	−0.1	−0.7 ^c	−0.4
Lung and bronchus	56.8	37.2	79.5	52.9	40.7	26.2
APC	−0.8 ^c	−0.2	−1.8 ^c	−0.6	+0.7 ^c	+0.7
Colon/rectum	21.2	14.7	25.8	18.5	18.0	12.1
APC	−1.7 ^c	+2.2	−2.0 ^c	+5.1	−1.7 ^c	−0.2
Prostate			32.9	21.9		
APC			−3.4 ^c	−3.8 ^c		
Breast					27.7	14.9
APC					−2.4 ^c	−0.6

SEER: Surveillance, Epidemiology, and End Results; AIAN: American Indians and Alaska Natives; APC: annual percent change.

^a Incidence data are from the 12 SEER areas (San Francisco, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, San Jose—Monterey, Los Angeles, and Alaska) for 1996–2000. Source: SEER Program public-use data (1973–2000), National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Cancer Statistics Branch, released April, 2003, based on the November, 2002 submission using bridged race categories (documented at <http://www.cdc.gov/nchs/about/major/dvs/popbridge/popbridge.htm>).

^b APC calculations were based on data from 1992–2000.

^c The APC differed significantly from zero ($P < 0.05$).

^d Death rate data were analyzed from a public-use file provided to SEER by the National Center for Health Statistics.

AIAN populations demonstrating no significant decrease during the same period.

Figures 2–7 break the total population into its constituent racial/ethnic groups for 1992–2000, as reported prior to the recent modifications announced by the United States Office of Management and Budget.¹⁰ The graphs utilize the joinpoint technique to analyze changes in trends. When we compare rate levels across the four cancers, it appears that the AIAN rates are among the lowest of the groups. However, if we look at the trends, then the AIAN population seems to be having a different cancer experience than other populations.

Figure 2 shows that female lung cancer incidence has a nearly flat trend for all groups except for Hispanic and AIAN women, who have significant downward slopes. The downturn at the end of the period for white women will require more years of data to determine significance. For AIAN women, the pattern has some variability, although the overall trend is downward. For males, the incidence rates are trending downward for all groups except the Asian/Pacific Is-

landers (API). The greater amount of variability in the male AIAN rate contributes to the lack of a significant trend, although a downward pattern emerges.

Lung cancer death rates (Fig. 3) among women appear fairly steady across the years. Although both white and black rates have been rising slowly but steadily, they may be flattening in most recent years. Among men, all racial/ethnic groups have falling rates, including AIAN men, who show a possible change of direction in 1998.

For colorectal cancer, incidence rates (Fig. 4) for women have remained comparatively steady, with the appearance of a slight downward trend for whites. Although there may be an increasing rate among the AIAN population, it is not statistically significant, and the most recent year is low. Among males, incidence has been decreasing or flat, except for Hispanics. Death rates are moving down for white, black, and API men and women, they remain steady for Hispanics and AIAN women, and they are rising for AIAN men (Fig. 5).

For prostate cancer (Fig. 6), higher incidence dur-

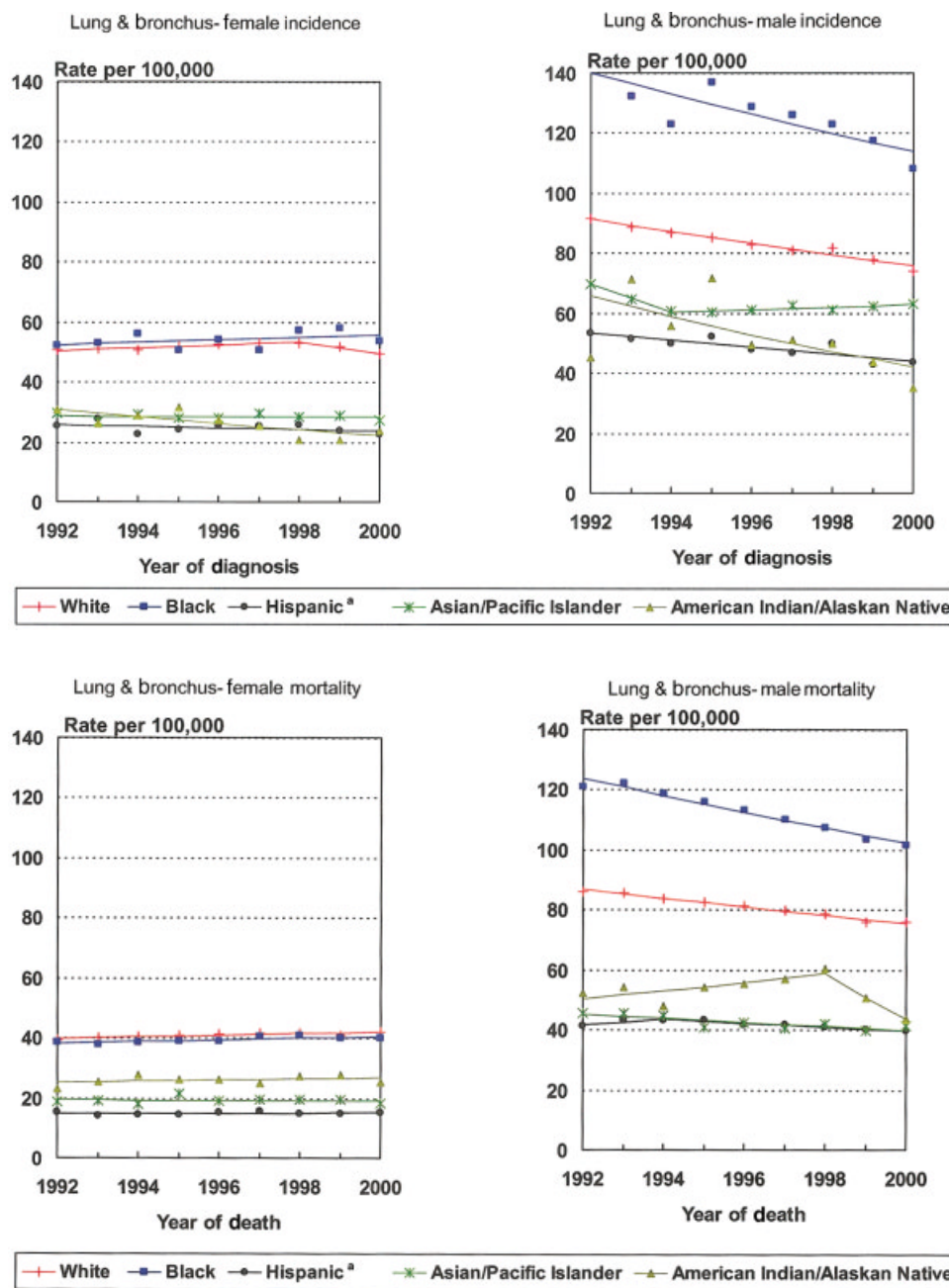


FIGURE 2. Lung and bronchus cancer: Surveillance, Epidemiology, and End Results (SEER) incidence, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: 12 SEER areas (San Francisco, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, San Jose—Monterey, Los Angeles, and Alaska Native registries); incidence data for Hispanics do not include cases from the Detroit, Hawaii, or Alaska Native registries. Regression lines were calculated using the Joinpoint Regression Program.

FIGURE 3. Lung and bronchus cancer: United States death rates, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: mortality data for Hispanics do not include data from Connecticut, Oklahoma, New York, or New Hampshire; mortality data for all other races are from all states in the United States. Regression lines were calculated using the Joinpoint Regression Program.

ing the period of increased awareness in the early 1990s was followed by a decrease until the middle of the decade, when rates began slowly rising, except for AIAN men. Deaths for all populations appear to be decreasing, with the downward trend for whites increasing in the middle of the decade, when an upward trend for Hispanics also turned downward. Looking at female breast cancer trends (Fig. 7), incidence rates appear to be increasing for all population groups except AIAN women and a recent downward trend among API women. The only significant increase was

among white women. Death rates have decreased significantly for whites, blacks and Hispanics over 1992–2000. The amount of decrease for white women is not shared by other groups; in fact, the death rates for AIAN women appear nearly flat.

With regard to cancer survival, a recent analysis of data from the nine SEER registries with the longest continuous period of data collection found some of the lowest survival rates for the AIAN population, including survival rates for all cancers combined.⁹ This study provided the first known population-based

FIGURE 4. Cancer of the colon/rectum: Surveillance, Epidemiology, and End Results (SEER) incidence, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: 12 SEER areas (San Francisco, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, San Jose—Monterey, Los Angeles, and Alaska Native registries); incidence data for Hispanics do not include data from the Detroit, Hawaii, or Alaska Native registries. Regression lines were calculated using the Joinpoint Regression Program.

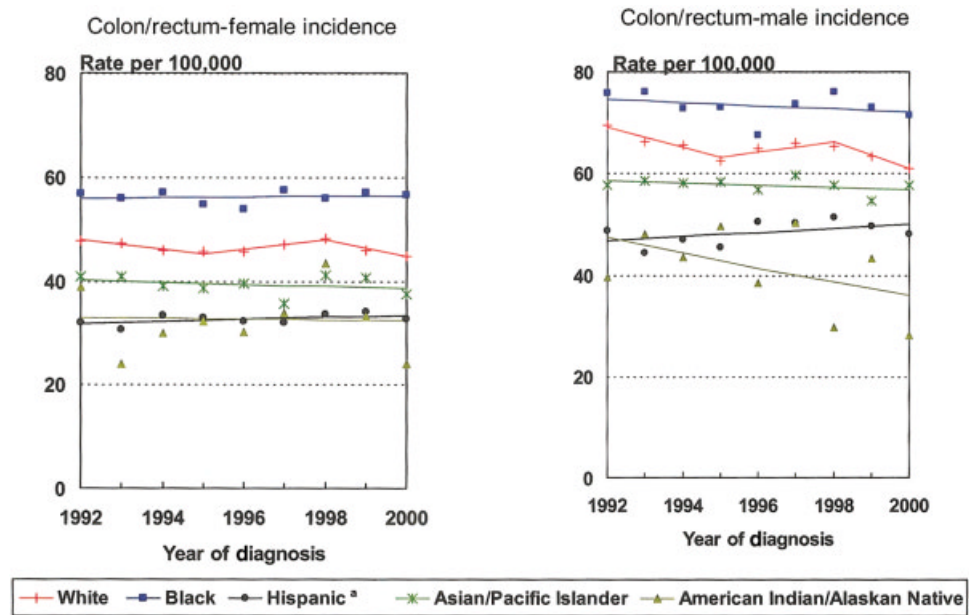
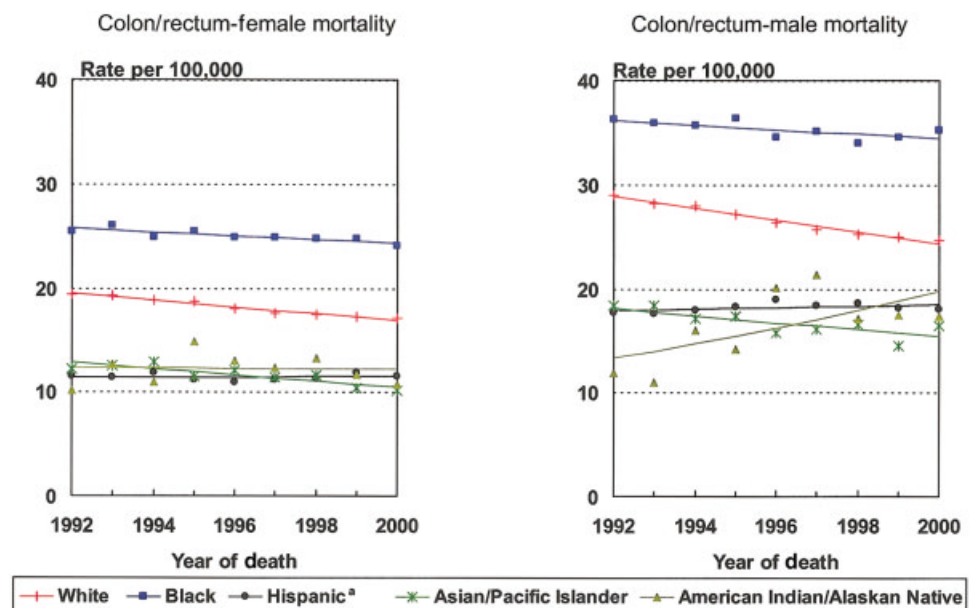


FIGURE 5. Cancer of the colon/rectum: United States death rates, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: mortality data for Hispanics do not include data from Connecticut, Oklahoma, New York, or New Hampshire; mortality data for all other races are from all states in the United States. Regression lines were calculated using the Joinpoint Regression Program.



comparisons of cancer-specific survival and the relative risk of cancer death among the six major racial/ethnic groups in the United States. The analysis shows a wider spread among races for cancers with more effective treatments—breast, prostate, and colorectal—and smaller differences for cancers with fewer options for treatment, such as lung cancer (Figs. 8, 9). Although adjusting for age and stage at diagnosis brings the curves closer together, a significant difference remains. Native Americans had the lowest survival rates for prostate, lung, and female colorectal

cancers and the second lowest survival rates for breast and male colorectal cancers.

DISCUSSION

When we consider the 12 SEER areas, we can see a slight decrease in cancer incidence and death rates for 1992–2000 overall for Americans. There remains an increasing incidence in women of 0.2% over the period. For the AIAN population within the SEER coverage areas, the levels of cancer may be lower, but the trends for several of the cancers studied appear to be

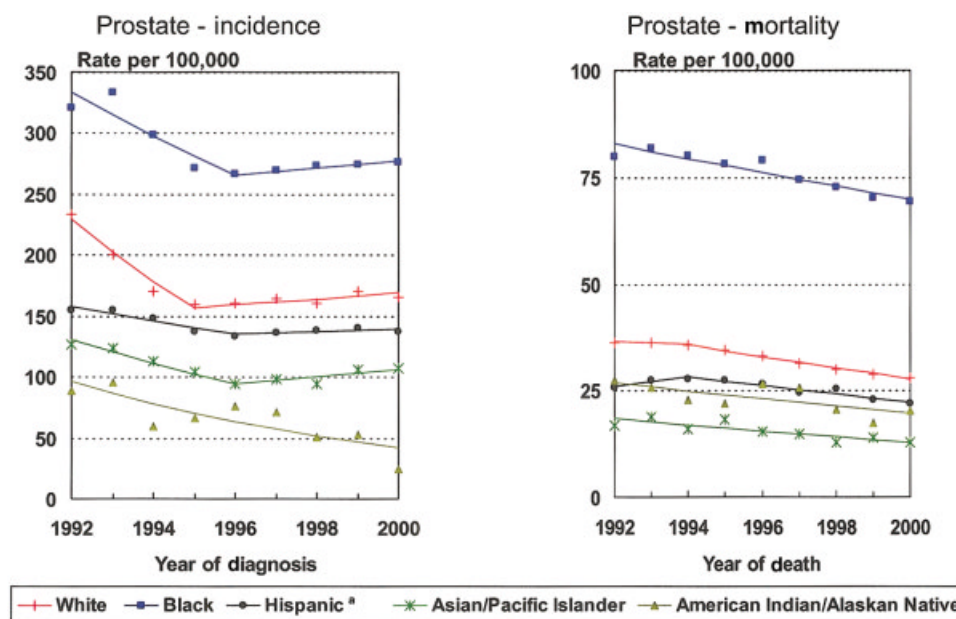


FIGURE 6. Prostate cancer: Surveillance, Epidemiology, and End Results (SEER) incidence and United States death rates, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: 12 SEER areas (San Francisco, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, San Jose—Monterey, Los Angeles, and Alaska Native registries); incidence data for Hispanics do not include cases from the Detroit, Hawaii, or Alaska Native registries; mortality data for Hispanics do not include data from Connecticut, Oklahoma, New York, or New Hampshire; mortality data for all other races are from all states in the United States. Regression lines were calculated using the Joinpoint Regression Program.

moving in a different direction than the other groups. The overall lack of a significant decrease in the death rate may be driven by increasing lung/bronchus cancer deaths among AIAN women and colorectal cancer deaths among AIAN men. (Not shown in the table is another contributor, liver cancer, which has a comparatively high rate among the AIAN population.³) Death rate from lung cancer is one of the 17 health status indicators developed for Healthy People 2010 to facilitate the comparison of populations at the national, state, and local levels, and lowering mortality from lung cancer and colorectal cancer is among the Healthy People 2010 objectives.^{3,11,12}

Examination of the joinpoint trend analyses of incidence and deaths reveals no significant changes for AIAN over the period, except for a decrease in the prostate cancer death rate. For breast cancer, we do not see the same picture that we see for other populations, in which increased screening has led to increasing incidence and subsequent decreasing mortality.^{13,14} New modeling techniques are being studied and may prove useful for interpreting rates that are highly variable from year to year, such as those for the AIAN population.

Beyond small population size, another difficulty encountered in establishing an overall rate for the

AIAN population is the substantial interregional differences in rates that were documented several years ago by the IHS.¹⁵ The IHS data show quite different patterns of mortality between the Southwest tribes and the Northern Plains tribes, with the latter tribes showing much higher death rates for lung, colon, and breast cancers. In addition, except for the Alaska Native Tumor Registry (ANTR), cancer registries do not report Alaska Natives separately from American Indians. The ANTR has been collecting and reporting cancer data for many years, issuing regular reports on cancer incidence,¹⁶ with an additional report on survival¹⁷ and, most recently, a 30-year report covering the period 1969–1998.¹⁸ Their reports indicate that Alaska Natives have some of the highest cancer rates, particularly for lung and colon cancers. These examples documenting regional differences add to the argument for increased local data collection and reporting.

The differences in cancer rates among races and between genders, as stated previously, may be a sign not only of differences in underlying risk but in detection and treatment as well, including both access to care and quality of care. These factors also complicate the study of cancer survival, which is subject to lead-time bias, length bias, and the differential effects of

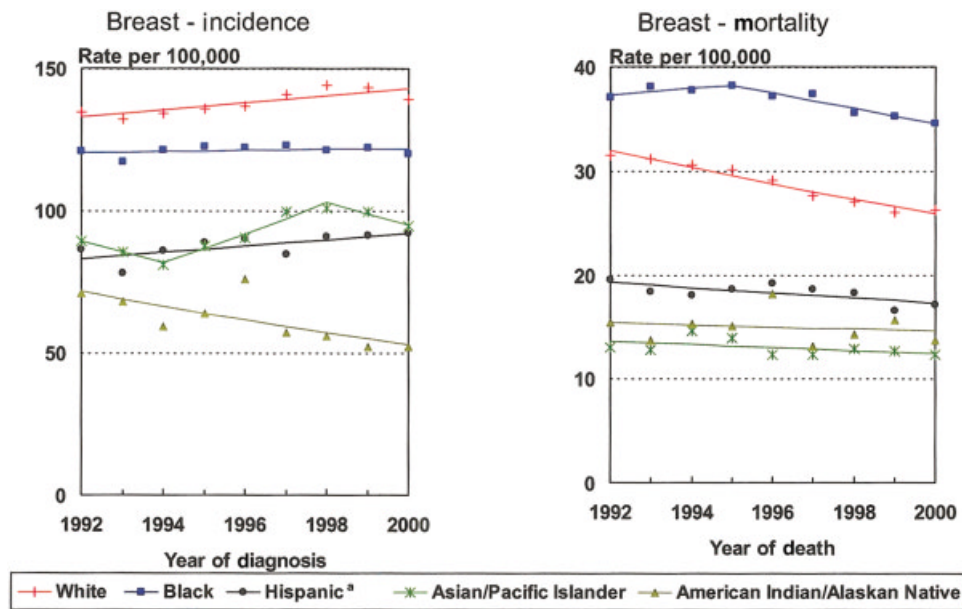


FIGURE 7. Female breast cancer: Surveillance, Epidemiology, and End Results (SEER) incidence and United States death rates, 1992–2000, by race/ethnicity. ^aCategorization as *Hispanic* is not mutually exclusive from categorization as *white*, *black*, *Asian/Pacific Islander*, or *American Indian/Alaskan Native*. Data source: 12 SEER areas (San Francisco, Connecticut, Detroit, Hawaii, Iowa, New Mexico, Seattle, Utah, Atlanta, San Jose—Monterey, Los Angeles, and Alaska Native registries); incidence data for Hispanics do not include cases from the Detroit, Hawaii, or Alaska Native registries; mortality data for Hispanics do not include data from Connecticut, Oklahoma, New York, or New Hampshire; mortality data for all other races are from all states in the United States. Regression lines were calculated using the Joinpoint Regression Program.

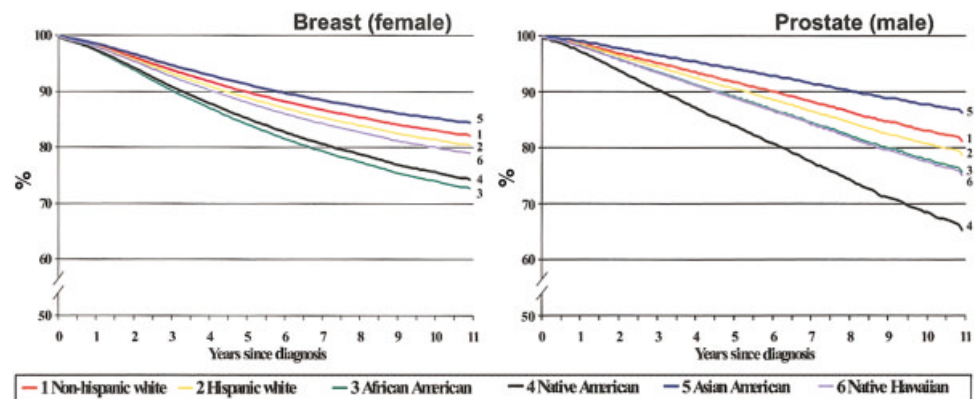


FIGURE 8. Cancer survival by race/ethnicity adjusted by age and stage of disease at the time of diagnosis from the Surveillance, Epidemiology, and End Results (SEER) Program, 1988–1997. Adapted from Clegg et al., 2002.⁹

interventions, such as screening.^{19–21} In addition, there are scant data available for the study period on small populations like Native Americans. Despite this, the survival study referenced herein extended and expanded previous studies that focused on whites and African Americans²² and holds potential for further expanded analyses with the addition of information from the newer registries.

Other Sources of Data

The limitations of using cancer registry data for Native Americans lead to a long list of caveats, which include

the lack of generalizability to areas not covered by registries and differential amounts of racial misclassification among existing data collection systems. However, there are important sources of additional information under development on Native Americans and cancer, some of which are expansions on and improvements to current efforts. Several of these are described below.

Expansion of registries

In February, 2001, the NCI announced the expansion of the SEER Program with awards to four additional

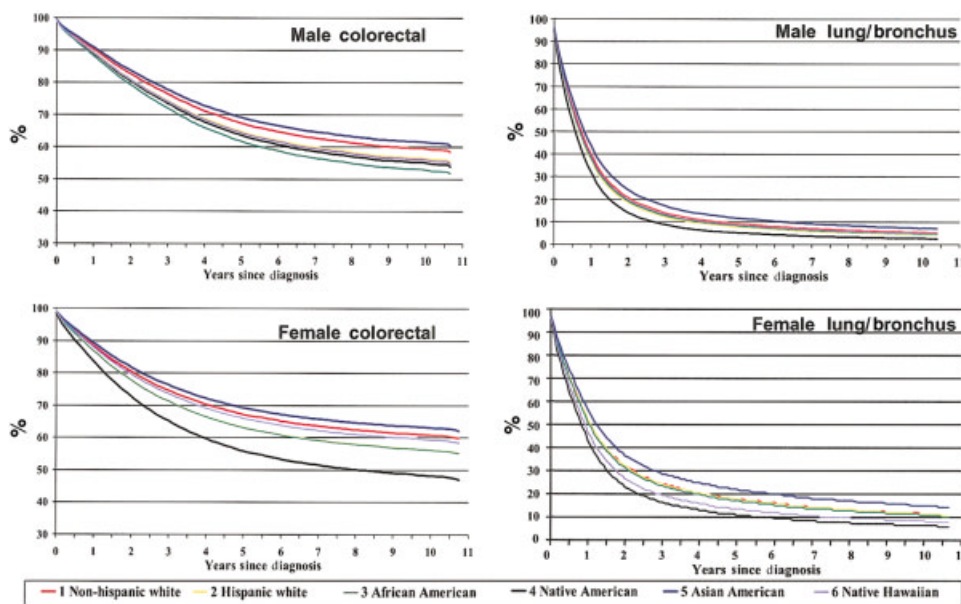


FIGURE 9. Cancer survival by race/ethnicity adjusted by age and stage of disease at the time of diagnosis from the Surveillance, Epidemiology, and End Results (SEER) Program, 1988–1997. Adapted from Clegg et al., 2002.⁹

states: Louisiana, Kentucky, New Jersey, and California (the remaining portions of the state not already within the SEER Program). The contracts were awarded through competitive selection and peer review, with emphasis on 1) coverage of populations for which limited data currently exist and 2) data quality measures available at the time of the procurement. Expansion to the 4 new areas increased SEER coverage to 23% of African Americans, 40% of Hispanics (32% of non-Mexican Hispanics), 42% of AIANs, 53% of Asians, and 70% of Native Hawaiians and Pacific Islanders. Data from the newest SEER registries will be reported for the first time in 2003, and, overall, SEER coverage will increase from 14% to 26% of the United States population (from about 39 million to nearly 74 million). The NCI funds for expansion of the SEER Program complement funding from the CDC through NPCR as well as funding from the states, resulting in enhanced coordination between the two agencies. The CDC supports 45 states, the District of Columbia, and 3 United States territories for enhancing established registries or for developing and implementing new registries.²³ Both the NCI and the CDC are working with the North American Association of Central Cancer Registries (NAACCR) to assist state registries in achieving data compatibility as they pool data to provide national estimates.

However, pooling data from different areas of the country may mask regional differences for AIAN populations, considering the wide variability of demographics, traditions, and resources. In the Southwest, through the SEER Program, the New Mexico Tumor Registry (NMTR) has been a reliable source of state-

wide cancer data since 1969, including data on American Indians.²⁴ The NMTR also collects data on Arizona Indians that is being prepared for publication. The NMTR has been providing technical assistance to smaller, regional cancer registries that cover specific AIAN groups. With this assistance, the Alaska Native Tumor Registry was able to reach the status of an official SEER registry. In addition, the NMTR is enabling the NCI to support pilot work for a cancer registry in partnership with the Cherokee Nation of Oklahoma.

Misclassification

Several studies using linkage of cancer registry data with the IHS and/or tribal records have documented that reported cancer rates for American Indians can be unreliable due to misclassification of individuals as non-Indian in the central registries. Findings have ranged from nearly negligible misclassification in the NMTR to as much as 50% in several state registries for the years studied.^{15,24–28} Currently, all SEER registries are participating in a data linkage project with the IHS. Conducted in a secure environment with the strictest confidentiality, this project is identifying additional cancer cases among American Indians. The results will be reported back to the registries, so their data bases can be updated. Although there are several limitations to this study, such as the fact that some AIANs have private insurance and are not included in the IHS files and that IHS coverage varies by geographic region, it has been a relatively cost-efficient first approach for updating the SEER data across the Nation. In an abstract presented at the June, 2002 meeting of NAACCR,

it was reported that 1094 American Indians with cancer had been misclassified as non-Indian, increasing the number of cases in SEER for the years 1988–1998 by 25.8%.²⁹ The success of this study has led to expanding the project to NPCR states with funding from the CDC.

In another effort, the NCI is supporting a data linkage project for the Northwest Portland Area Indian Health Board, which also is addressing the issue of misclassification and developing software that can be used by other registries to improve reporting of Native American status.³⁰ In this case, the linkage is improving case identification by using tribal information.

Ecologic studies

If we want to consider cancer rates by factors other than race/ethnicity, then we can look to recent studies in which cancer death rates were examined by various demographic and socioeconomic status (SES) determinants in ecologic analyses.^{31,32} Using census information for the decades covered to classify all counties in the United States by quintiles of these variables from low to high, these analyses point to a differential burden of cancer. They also reveal how the gradient among SES groups can change; for example, among men, lung cancer shows a crossover in death rates since the 1950s, when the highest SES quintiles had the highest rates. A new software tool for ecologic analysis soon will be available. There is an ongoing debate over whether any more detailed levels of data should be made available for analysis, particularly they are linked to geographic location, but the potential usefulness of a geospatial, ecologic analysis of AIAN cancers makes a compelling argument.

Conclusions

Beyond the NIH, many organizations have been developing strategic initiatives to address health disparities, including the American Medical Association and the World Health Organization, and the list is growing constantly. The application of the results of these efforts will rely on accurate data. In the current report, we have demonstrated that this is a complex problem among AIAN populations, that regional differences tell us the data cannot be pooled easily, and that improving local data collection and reporting may provide greater benefit to tribal communities.

On a positive note, there are more data available today than ever before on cancer among AIAN populations, and the network of cancer registries is expanding. In the 2002 *Annual Report to the Nation*, incidence data that met a specified standard were reported for approximately 55% of the United States population using combined SEER and NPCR data.¹² It

is still true that more data are needed, particularly to facilitate comparisons for all AIAN groups whose rates and risks vary so much by geographic region. Nevertheless, more data are being collected that can be analyzed in more ways than were available previously to describe and define the cancer burden more precisely than was possible in the past.

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